Thermodynamics and Statistical Physics

1st Exercise Class

Temperature Scales, Heat Expansion

Exercises for the Class

1. The length of a mercury column in a mercury-in-glass thermometer is 30 mm at ice point and 180 mm at steam point. What is the temperature when the length is 60 mm?

2. Inside the wall of a house, an L-shaped section of hot-water pipe consists of three parts: a straight, horizontal piece $h = 28.0$ cm long; an elbow; and a straight, vertical piece, $\ell = 134$ cm long (Fig. 1). A stud and a second-story floorboard hold the ends of this section of copper pipe stationary. Find the magnitude and direction of the displacement of the pipe elbow when the water flow is turned on, raising the temperature of the pipe from 18.1 °C to 46.6 °C. The linear expansion coefficient for copper is $\alpha = 1.7 \cdot 10^{-5}$ 1/°C.

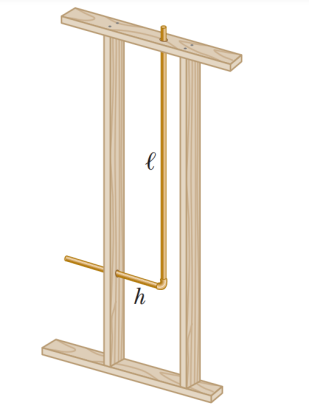


Fig. 1

3. A concrete road surface consists of 15 m long sections separated by gaps to allow for thermal expansion as the weather changes. Suppose the expected road surface temperatures vary during the year from a low of −15 °C to a high of 45 °C. (Note: On a sunny day, road surface temperatures can be much higher than air temperatures.) The road is designed so that there remains a 2.0 mm gap between sections on the hottest day. What is the gap on the coldest day? The linear expansion coefficient for concrete is $\alpha = 1.2 \cdot 10^{-5}$ 1/°C.

4. Show that for small thermal expansions of a solid object the linear and volume coefficients of expansion are related by $\beta \approx 3\alpha$.

5. A mercury thermometer is constructed as shown in Fig. 2. The glass capillary tube has a diameter of 0.004 cm, and the bulb has a diameter of 0.25 cm. Find the change in height of the mercury column that occurs with a temperature change of 30 °C. ($\beta_{\text{Hg}} = 1.8 \cdot 10^{-4}$ 1/°C, $\alpha_{\text{glass}} = 4.0 \cdot 10^{-6}$ 1/°C)

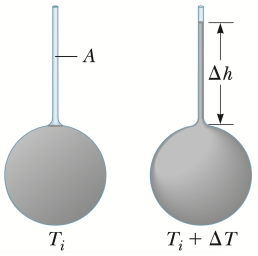


Fig. 2

6. A square hole 8.00 cm along each side is cut in a sheet of copper.

   (a) Calculate the change in the area of this hole resulting when the temperature of the sheet is increased by 50.0 K.

   (b) Does this change represent an increase or a decrease in the area enclosed by the hole?

7. A bimetallic strip of length L is made of two ribbons of different metals bonded together.

   (a) First assume the strip is originally straight. As the strip is warmed, the metal with the greater average coefficient of expansion expands more than the other, forcing the strip into an arc with the outer radius having a greater circumference. Derive an expression for the angle of bending α as a function of the initial length of the strips, their average coefficients of linear expansion, the change in temperature, and the separation of the centers of the strips.

   (b) Show that the angle of bending decreases to zero when ΔT decreases to zero and also when the two average coefficients of expansion become equal.

   (c) What happens if the strip is cooled?

8. Consider a wire clamped between two rigid supports and thus held at constant length L. Suppose, the wire is now cooled from T_1 to T_2, calculate the resulting change of the force, which stretches the wire. Note that the equilibrium states of the wire are fixed by specifying two of the state variables F (stretching force), L (length), and T (temperature, which are related by some equation of state

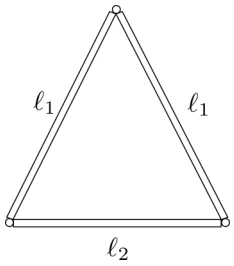
$$g(F, L, T) = 0.$$

Therefore, the so-called, cyclical relation can be used:

$$\left(\frac{\partial F}{\partial T}\right)_L \left(\frac{\partial L}{\partial F}\right)_T \left(\frac{\partial T}{\partial L}\right)_F = -1.$$

Homework to Be Submitted

1. The legs of the isosceles triangle are made of material with thermal expansion α_1, and its base is made of material with thermal expansion α_2. The ratio of the length of the base to the length of the legs is $\ell_2/\ell_1 = n$. At the vertices of the triangle, the rods are connected to each other with hinges. Determine the relationship between α_1 and α_2 if the distance of the system's center of mass from the apex of the triangle does not change due to temperature change.



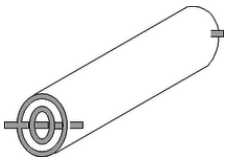
Homework for Practice

1. A liquid has a density ϱ.
   (a) Show that the fractional change in density for a change in temperature ΔT is
   $$\Delta\varrho/\varrho = -\beta\Delta T.$$
   (b) What does the negative sign signify?
   (c) Fresh water has a maximum density of 1.0000 g/cm^3 at 4.0 °C. At 10.0 °C, its density is 0.9997 g/cm^3. What is β for water over this temperature interval?
   (d) At 0 °C, the density of water is 0.9999 g/cm^3. What is the value for β over the temperature range 0 °C to 4.00 °C?

2. Following a collision in outer space, a copper disk at 850 °C is rotating about its axis with an angular speed of 25.0 rad/s. As the disk radiates infrared light, its temperature falls to 20 °C. No external torque acts on the disk.
   (a) Explain how the angular speed changes as the disk cools.
   (b) What is its angular speed at the lower temperature?

3. A 1.0 km steel railroad rail is fastened securely at both ends when the temperature is 20 °C. As the temperature increases, the rail buckles, taking the shape of an arc of a vertical circle. Find the height h of the center of the rail when the temperature is 25 °C.

4. We insert a thinner pipe into a pipe and fasten both ends together. The pipes fit loosely, are of equal length, but are made of different materials.



   What is the linear thermal expansion coefficient along the length of the double pipe formed in this way?

   The cross-sectional areas of the pipe walls are A_1 and A_2, and the linear thermal expansion coefficients and Young's moduli of the two different materials are α_1 and α_2, and E_1 and E_2, respectively.